🛠️ Tech Stack

text2sqlAgent - Voice-Enabled Multilingual NL-to-SQL System

⚙️ Backend

Technology	Purpose	Version
Python	Core programming language	`3.8+`
Flask	Web framework / REST API	`≥2.3.0`
Flask-CORS	Cross-origin request handling	`≥4.0.0`
SQLite	Database (built-in Python)	`Built-in`

🤖 AI / Machine Learning

Technology	Purpose	Version
Google Gemini 2.0 Flash	Online LLM for SQL generation	`google-generativeai ≥0.3.0`
Ollama (TinyLlama 1.1B)	Offline LLM alternative	`Local server`
Sentence Transformers	Text embeddings (paraphrase-multilingual-MiniLM-L12-v2)	`≥2.2.0`
FAISS	Vector similarity search (384-dim)	`faiss-cpu ≥1.7.0`

📊 Data Processing

Technology	Purpose	Version
Pandas	Data manipulation	`≥1.5.0`
NumPy	Numerical computing	`≥1.21.0`

📈 Visualization

Technology	Purpose	Version
Plotly	Interactive charts (bar, line, pie, scatter)	`≥5.15.0`
Matplotlib	Static chart generation	`≥3.6.0`
Seaborn	Statistical visualizations	`≥0.11.0`

🌐 Frontend

Technology	Purpose
HTML5 / CSS3 / JavaScript	Web UI (templates/index.html)
Web Speech API	Voice input (Telugu, Hindi, English)

Plotly.js	Chart rendering in browser

🔧 Other Tools

Technology	Purpose
python-dotenv	Environment variable management
Requests	HTTP client (Ollama API calls)
Logging	System monitoring

🏗️ Architecture Diagram

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│                      Frontend (HTML/JS)                      │
│          Voice Input (Web Speech API) + Query Form           │
└──────────────────────┬──────────────────────────────────────┘
                       │ HTTP REST API
┌──────────────────────▼──────────────────────────────────────┐
│                    Flask Web Server                          │
│                      (main.py)                               │
├──────────────────────────────────────────────────────────────┤
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐       │
│  │   FAISS      │  │  Gemini/     │  │   SQLite     │       │
│  │  Similarity  │  │  Ollama LLM  │  │  Database    │       │
│  │   Index      │  │              │  │              │       │
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│                  │   Plotly     │                            │
│                  │ Visualization│                            │
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🎯 Key Highlights

🌐 Multilingual

Supports English, Telugu, Hindi via multilingual embeddings

🔌 Hybrid AI

Online (Gemini) + Offline (Ollama) modes

⚡ Smart Caching

FAISS similarity matching reduces LLM calls by ~70%

📊 Auto-Visualization

Automatically detects chart type from query results